

Experiment 9

1. Experiment Title

Implement LocalStorage and SessionStorage to Save User Preferences

2. Software/Tools Required

- Visual Studio Code
- Google Chrome / Microsoft Edge
- HTML5
- CSS3
- JavaScript
- Live Server Extension

3. Theory

5.1 LocalStorage

`localStorage` is a browser storage mechanism used to store data as key-value pairs.

Data stored in LocalStorage remains available even after the browser is closed and reopened.

Example:

```
localStorage.setItem("theme", "dark");
```

To retrieve the stored value:

```
localStorage.getItem("theme");
```

To remove a value:

```
localStorage.removeItem("theme");
```

5.2 SessionStorage

`sessionStorage` is also used to store key-value data, but the stored data is associated with the current browser session.

To retrieve it:

```
sessionStorage.getItem("username");
```

To remove it:

```
sessionStorage.removeItem("username");
```

5.4 Saving User Preferences

The experiment uses LocalStorage to remember the user's selected **Light/Dark theme**.

```
localStorage.setItem("theme", theme);
```

When the page loads, the saved theme is retrieved:

```
const savedTheme =  
    localStorage.getItem("theme");
```

This allows the application to automatically restore the user's preference.

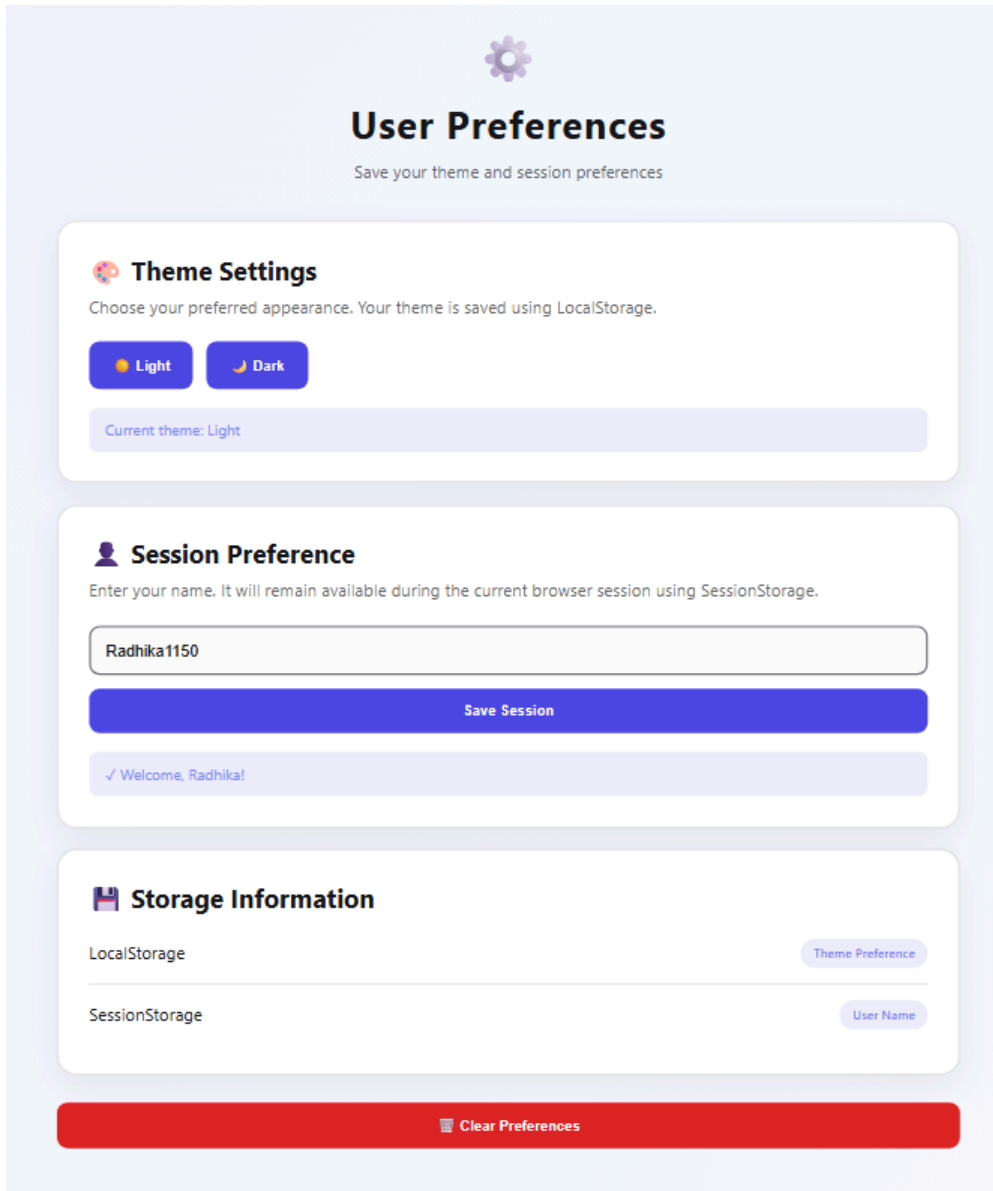
5.5 Restoring Session Data

The username is saved in SessionStorage:

```
sessionStorage.setItem(  
    "username",  
    username  
);
```

When the page is loaded again during the same session, the username is retrieved and displayed.

4. Output



The screenshot displays a 'User Preferences' interface with a light blue background and a purple gear icon at the top center. The title 'User Preferences' is in bold black text, with the subtitle 'Save your theme and session preferences' below it. The interface is divided into three main sections: 'Theme Settings', 'Session Preference', and 'Storage Information'. The 'Theme Settings' section includes a 'Theme Settings' header with a paint palette icon, a description 'Choose your preferred appearance. Your theme is saved using LocalStorage.', two buttons labeled 'Light' and 'Dark', and a status bar showing 'Current theme: Light'. The 'Session Preference' section has a 'Session Preference' header with a person icon, a description 'Enter your name. It will remain available during the current browser session using SessionStorage.', a text input field containing 'Radhika1150', a blue 'Save Session' button, and a feedback bar showing '✓ Welcome, Radhika!'. The 'Storage Information' section features a 'Storage Information' header with a folder icon, a table with two rows: 'LocalStorage' (with a 'Theme Preference' tag) and 'SessionStorage' (with a 'User Name' tag). At the bottom, there is a red button labeled 'Clear Preferences' with a trash can icon.

User Preferences
Save your theme and session preferences

Theme Settings
Choose your preferred appearance. Your theme is saved using LocalStorage.

Light Dark

Current theme: Light

Session Preference
Enter your name. It will remain available during the current browser session using SessionStorage.

Radhika1150

Save Session

✓ Welcome, Radhika!

Storage Information

LocalStorage	Theme Preference
SessionStorage	User Name

Clear Preferences

5. Result / Conclusion

The **Gym Admission Form** was successfully designed and implemented using HTML, CSS, and JavaScript. JavaScript DOM methods were used to access form fields, while `input`, `change`, `blur`, and `submit` events were used for real-time validation. Regular expressions were used for email and phone-number validation. The system provides immediate feedback and prevents submission when invalid information is entered.